Purpose:
The purpose of this document is to provide guide line for SMT defects troubleshooting and which helps to restore the quality requirement standards.

Scope:
Surface Mount Technology (Solder paste printing, Pick & Place, Reflow & PCB).

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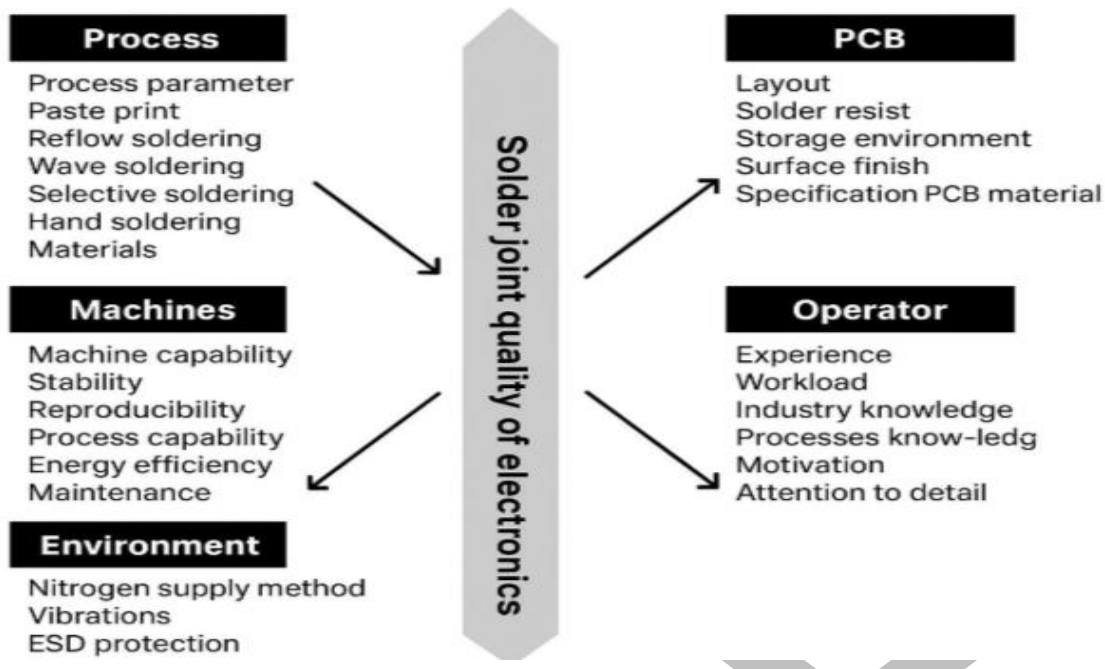
Review	Name
Prepared By	Stanish Vijay
Reviewed By	Sudhakar Babu
Approved By	Aditi Sharma

1. Quality Factors in Electronics Assembly:

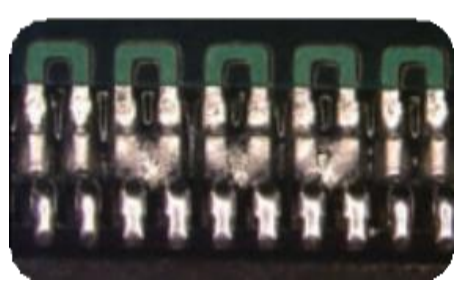


1.a) Measurement of the Profile with a reference board:

1. Reflow Profile Measurement with a Reference Board
2. Using a Reference PCB to Validate the Reflow Process
3. Determining the Reflow Temperature Curve via Reference Board Data

2. Bridging:

Defect Definition: Solder connecting, in most cases, miscon- necting two or more adjacent pads that come into contact to form a conductive path.



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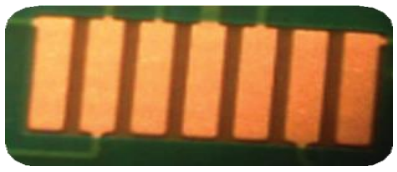
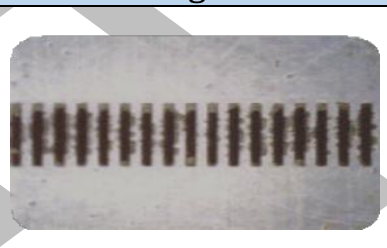
2.a) Bridging Possible Causes: PCB

Description	Recommendations	Image
Pads will contribute to coplanarity issue resulting in poor gasketing during printer setup.	Highly recommended to remove solder mask between adjacent pads especially for fine-pitch components (non-solder mask defined pads).	

2.b) Possible Causes Bridging (Stencil):

Description	Recommendations	Image
Dirty stencil with paste underneath will contaminate the bare board on the next print, attributing a potential bridge.	•Verify zero print gap set up. •Ensure minimum print pressure. •Increase wipe frequency. •Use different stencil cleaning chemistry.	

2.c) Possible Causes Bridging (Stencil):

Stencil tension:

Ensure stencil tension is tight. Poor stencil tension will make it impossible to have a good setup for consistent print definition.

Image for reference only

2.d) Bridging (Stencil Aperture Design):

For fine pitch component, it is highly recommended to have the opening slightly smaller than landing pad size to improve stencil to PCB gasketing.

2.e) Bridging (Defect Possible Causes in Solder Paste Printer):

Description	Recommendations	Image

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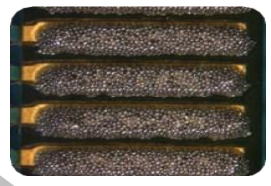
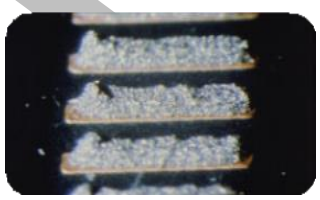
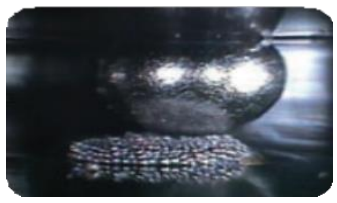
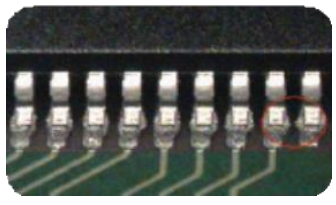
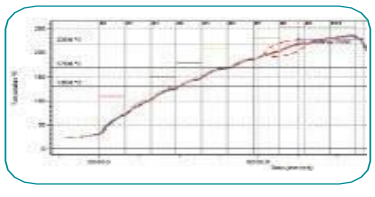
Poor gasketing – paste oozes out beneath stencil during printing, increasing chance of wet solder paste bridges.	1. Zero print gap between stencil and PCB 2. Check paste smear underneath stencil. 3. Check sufficient stencil tension.	
Misaligned print will challenge the paste to pull back to pads during molten stage, increasing the potential for bridging.	Ensure print accuracy and consistency for both print strokes.	
Smearing and bridging phenomenon on the next printed board after stencil cleaning operation	Verify stencil is dry after cleaning and before next print. Standard cleaning mode is wet/vacuum/dry.	
Poor print definition with dog ears especially on fine-pitch components	Check board support. Adjust separation speed to achieve minimum dog ears. NB: Different paste chemistry requires different separation speed to minimize dog ears.	
Dented squeegee blades could result in uneven print pressure.	Check squeegee blades condition.	

2.f) Possible Causes While Component Placement (Pick & Placement):

Description	Recommendations	Image
Placement inaccuracy will narrow the gap between pads, increasing the chance of bridging.	Recommendations Verify component placement pressure. Use X-ray to verify BGA placement. Use microscope for QFPs.	
Excessive component placement pressure will squeeze paste out of pads.	Verify actual component height against data entered in the machine Component placement height should be ±1/3 of paste height.	

2.g) Possible Causes in Reflow Profile:

Description	Recommendations	Image
Extended soak will input more heat to the paste and result in paste hot slump phenomenon.	• Adopt a straight ramp to spike profile, without soak zone if possible. • Be sure to check for BGA voiding when converting from a soak to a straight ramp profile.	

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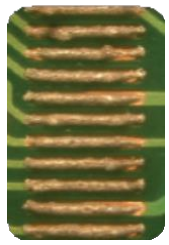
2.f) Possible Causes: Solder Paste:

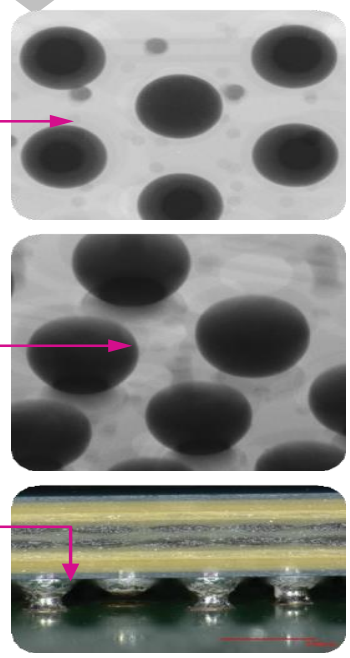
Description	Recommendations	Image
Dry paste phenomenon – irregular print shape and inconsistent print volume	Paste has expired or been exposed to excessive heat. Operating temperature within supplier's recommendations. Check temperature inside printer. Normal requirement around 25°C, 50%RH	
Paste oozes out of pads, may form connection with adjacent pads.	Do not mix using new and old paste. Operating temperature within supplier's recommendations Verify with another batch of paste to confirm problem is batch-related. Perform cold and hot slump test result using IPC-TM-650 Method 2.4.35.	

3. Non-Wet Opens:

3.a) Non-Wet Defect Definition:

A Non-Wet Open (NWO) defect, also known as non-wet or lifted ball, occurs during the Surface Mount Technology (SMT) assembly reflow process when the solder sphere and solder paste on the BGA have no physical contact with the pad after reflow, yet there was paste on the pad prior to the board entering the oven. Conventional inspection techniques may not detect these defects, which can be identified by the presence of the non-wetted pad after reflow.

Non-Wet Opens Defect



Possible Causes from PCB:

Description	Recommendations
Poor PCB Finish	Adopt better quality metal surface finish such as higher Temperature Resistant OSP or ENIG.

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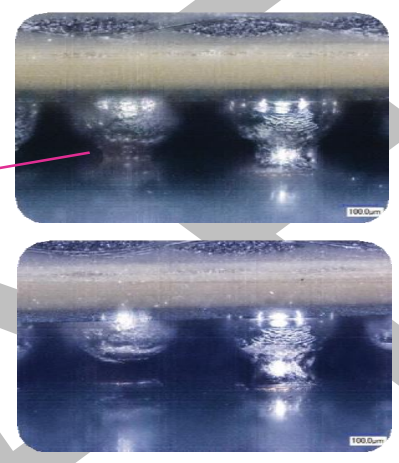
Components that have a very high warping signature characteristic can lead to NWO defects. During reflow, the solder paste can have a higher propensity to adhere to the solder sphere vs. the pad, especially on OSP coated boards. This happens when the component warps and the distance from the component pad becomes so great that the solder paste is physically lifted off the pad and then reflows onto the ball during the time above liquidus (TAL), thus not leaving any solder paste on the pad.	Using a low-temperature solder paste in applications where temperature sensitive components are involved can help to avoid Non-Wet Opens (NWO) and other defects. Low temperature soldering is also ideal where higher reflow temperatures may cause failures.

Image before and after cleaning flux off the assembly



Possible Causes from Reflow Profile:

Description	Recommendations
Profile	Profile Optimization

3.b) Possible Causes from Stencil:

Description	Recommendations
Insufficient volume of solder at the point of the non-wet open where there is not enough flux left to overcome the OSP once the part comes back from warping.	Optimize the stencil design by increasing the volume of paste by opening up the stencil apertures to provide more paste volume on the pad. Increase the aperture-to-pad ratio to 1:1 or 1:1.1 or based on component warpage.

Possible Causes from Solder Paste:

Description	Recommendations

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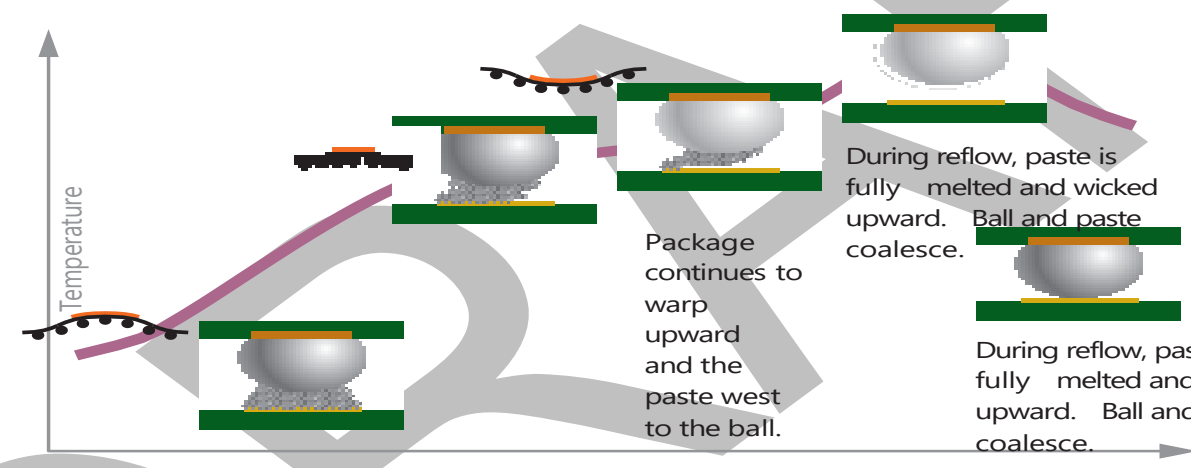
Many popular commercially available solder pastes have a greater propensity to adhere to the sphere vs. the OSP pad during reflow. Along with the high warping signature of these components, this creates a condition in which the paste is no longer in contact with the pad and thus reflows to the ball.	Switch to a solder paste with chemistry specifically designed to be NWO- resistant. A solder paste that has the proper wetting force and time balance characteristics to overcome the paste lifting, but still has enough activity to overcome the OSP on the pad when the part comes out of the warping stage and flattens out is ideal. There are a number of solder paste formulations available that have been designed and/or have been tested against this defect.

3.c) Non-Wet Open:

Non-Wet Open Defect Formation Mechanism During soak, package-MB move away from each other due to warpage shape change. Paste starts to lift up from PCB land at ~150-180°C.

Solder paste preferentially wets to the solder ball rather than the PCB land. Mechanism primarily driven by paste characteristics.



Solder paste preferentially wets to the solder ball rather than the PCB land. Mechanism primarily driven by paste characteristics.

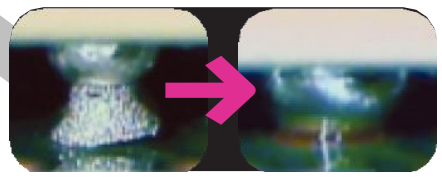
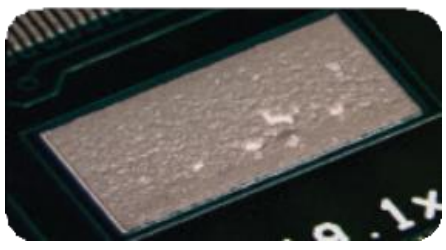


Illustration and image from Intel paper presented at SMTAI: "Fundamentals of the Non-Wet Open BGA Solder Joint Defect Formation,

4. Insufficient Solder paste Fill:

Definition: Amount of solder paste deposited on PWB at printer station is much less than stencil opening design.



4. a) Possible Causes from Stencil:

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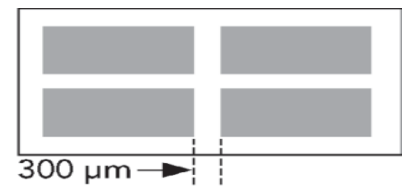
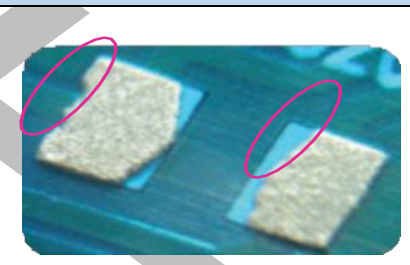
Description	Recommendations	Image
Paste scooping effect especially on large pads	Segment the large opening into smaller apertures. Check for excessive squeegee pressure	

4.b) Possible Causes from Solder Paste Printing:

Description	Recommendations	Image
Paste does not roll into aperture	Reduce print speed. Adopt lower squeegee contact print. Ensure paste is not expired or dry. Ensure sufficient board support. Reduce squeegee pressure.	

5. Insufficient Solder:

Definition:

Amount of solder paste deposited on PWB at printer station is much less than stencil opening design or, after reflow, insufficient solder to form a fillet at the component leads.

5.a) Possible Causes from Stencil:

Description	Recommendations
Solder paste adheres on the stencil aperture walls.	Area ratio > 0.59 for type 4 powder Aspect ratio > 1.5 No burr on stencil aperture edge.

5.b) Possible Causes on Solder paste Printing:

Description	Recommendations
Print definitions	Verify print setup Reduce print speed to provide sufficient time for paste to roll into aperture. Check stencil snap off speed vs. TB recommendation.

5.c) Possible Causes form Reflow:

Description	Recommendations

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Mismatch in CTE between component and PCB. can cause solder wicking effect which may look like insufficient solder on pads.	Attach thermocouple on component and PCB. Apply soak profile to minimize delta T before reflow zone. Set bottom zones to be higher temperature, if possible, to keep PCB hotter than component leads.

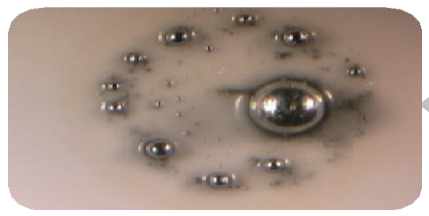
5.d) Possible Causes: Solder:

Description	Recommendations
Solder paste viscosity	Check paste conditions such as dry paste phenomenon by verifying if paste rolls or skids along print direction.

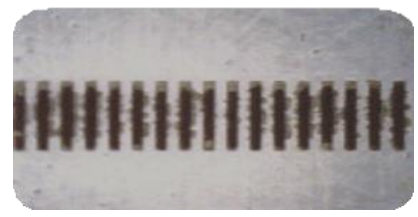
6. Random Solder Balls:

Definition:

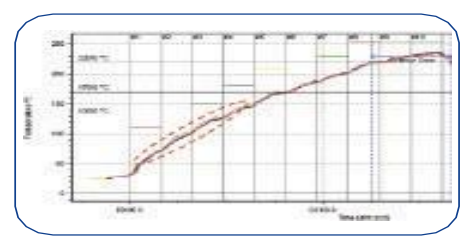
After reflow, small spherical particles with various diameters are formed away from the main solder pool.



6.a) Possible Causes form Stencil:

Description	Recommendations	Image
Paste stuck under the stencil will be transferred onto the solder mask of the next PCB.	Verify zero print gap set up. Check minimum print pressure used. Check cleaning efficiency such as wet/dry/vacuum. Check wipe frequency	

6.b) Possible Causes (Reflow):

Description	Recommendations	Image
Fast ramp-up rate or preheat rate will not allow sufficient time for the solvent to vaporize off gradually.	Slow preheat rate is recommended, typically < 1.5°C/sec from room temperature to 150°C.	

6.c) Possible Causes: PCB Moisture

Description	Recommendations
Trapped moisture may result in explosive vaporization.	Especially for lower grade PCBs such as FR2, CEM1, tends to absorb moisture. Bake 120°C for 4 hours if necessary.

6.d) Possible Causes: Solder Paste

Description	Recommendations	Image

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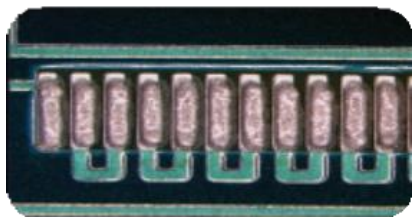
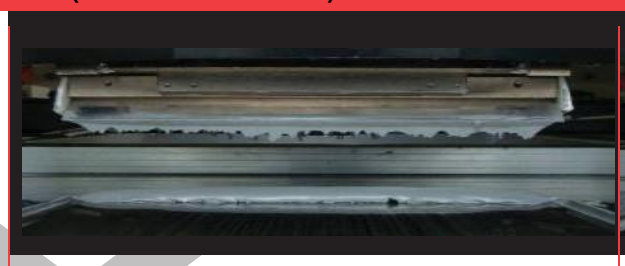
Especially for water-soluble solder paste which is hygroscopic, it tends to have limited stencil life because of moisture absorption	Minimize exposure time Printer temperature and humidity to be within recommendation Try new lot of solder paste to verify paste integrity. Use coarser powder size, if possible, as fine powder size has more oxides and tends to slump more readily.	

7. Paste Drying/Sticking to Squeegee:

Definition: When the printer's squeegee is lifted off the stencil between print strokes, paste should remain primarily on the stencil making it available for the next printing stroke.

7.a) Possible Causes:

Description	Recommendations
Paste bead too high	Maintain the solder bead height between 1.5 and 2 cm
Bead diameter too large	Use a taller squeegee blade
Paste replenishment not frequent enough	Over time with high component density assemblies, the ratio of metal to flux will increase
Printing water soluble paste in extremely low humidity	Better climate control when using water soluble paste

	T-0 (After 10 Kneads)	T-8 Hours
Lot# OM-5002 302042	 Temperature: 28.3°C Humidity: 48% RH	 Temperature: 32.2° C Humidity: 30% RH
Lot# OM-5002 302038	 Temperature: 25.8° C Humidity: 50% RH	 Temperature: 24.4° C Humidity: 38% RH

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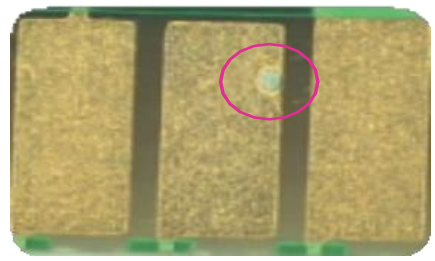
8. Solder Spattering:

Definition: Solder Spatter phenomenon is very similar to solder balling, but the concern is usually about solder deposited onto Au fingers.



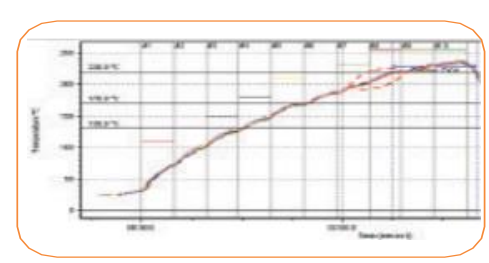
8.a) Possible Causes: PCB

Description	Recommendations	Image
Handling of boards	Do not mix clean and washed boards. Open fresh PCBs from package when ready to run. Ensure working area is cleaned thoroughly and not	
Bare boards contamination	Inspect bare PCBs to capture and filter solder found on bare PCB before printing station.	

8.b) Possible Causes: Screen

Description	Recommendations
Ineffective cleaning of stencil wipe will transfer small particles of solder onto the top surface of the next bare board.	Ensure wipe frequency is set correctly. Use effective solvent, preferably SC10. Use printer machine camera to inspect the effectiveness of stencil cleaning.

8:c) Possible Causes: Reflow

Description	Recommendations	Image
Control the flux out-gassing rate to minimize explosive solder scatter on Au pads.	For SAC 305, set slow ramp rate of 0.3-0.4°C/sec from 217-221°C.	

9. Mid-Chip Solder Balls (MCSB)

Definition: After reflow, large solder ball(s) is/are located on the side of the chip components, between the terminations and away from the pads.

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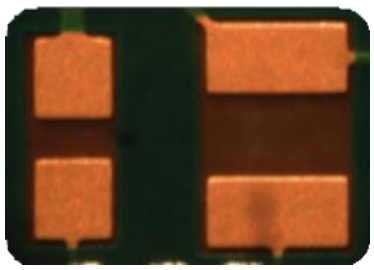
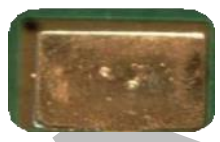
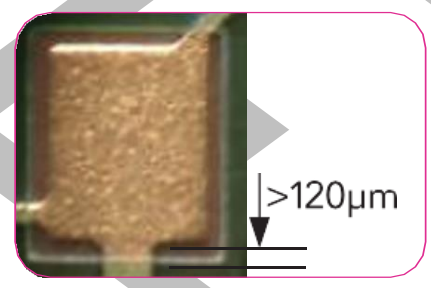
9.a) Possible Causes: PCB

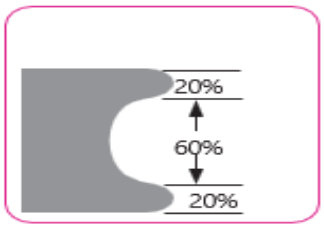
Description	Recommendations
Solder dissociation and does not adhere on solder mask	Remove solder mask between pads. Gap between pad and solder mask is recommended to maintain at least 75µm~100µm, preferably >120µm. Solder mask may not be centralized around pad.







9.b) Possible Causes: Stencil

Description	Recommendations	Image
Excess paste squeezes underneath component body tends to dissociate with the main body of solder during reflow.	Home plate or U-shape design may help to reduce the amount of paste potentially squeezed under the component body, onto the mask.	
	NB: Aperture reduction may not be suitable for component size smaller than 0603. Besides, LF alloy has higher surface tension and does not pull back after reflow.	

9.c) Mid-Chip Solder Balls (MCSB):

Definition: After reflow, large solder ball(s) is/are located on the side of the chip components, between the terminations and away from the pads.

9.d) Possible Causes: Solder Paste Printing:

Description	Recommendations
Paste smearing on solder mask	Printer set up for zero print gap, verified by paste height consistency without dog ears Print alignment accuracy

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9.e) Possible Causes: Component Placement

Description	Recommendations
Excessive placement pressure will squeeze paste on pad	Verify actual component height against data entered in the machine. Component placement height should be ±1/3 of paste height.

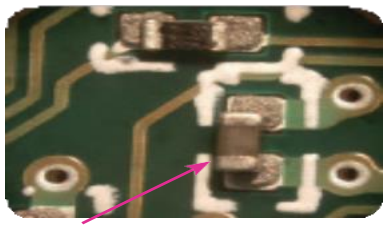
9.f) Possible Causes: Reflow

Description	Recommendations
Extended soak will input more heat to the paste and result in paste slump phenomenon.	Adopt a straight ramp to spike profile, without soak zone if possible. Be sure to check for BGA voiding when converting from a soak to a straight ramp profile

10.Tombstoning

Definition: A tombstone, sometimes called Manhattan effect, is a chip component that has partially or completely lifted off one end of the surface of the pad.

10.a) Possible Causes: Pad

Description	Recommendations	Image
Component body must cover at least 50% of both pads	If component terminations are not covering >50% of pads, high tendency to have imbalance wetting force, resulting in tombstoning. Feedback to supplier for alteration if possible.	
Unequal pad size especially with ground pad on one side	Unequal size means different solder volume, increasing potential for unequal wetting force. If due to design limitation, use a gradual soak ramp rate just before reaching liquidus point, e.g., SAC305, soak @ 190-220°C for 30-45	

10.b) Possible Causes: Placement Accuracy & Pressure

Description	Recommendations	Image

Review	Name
Prepared By	Stanish Vijay
Reviewed By	Sudhakar Babu
Approved By	Aditi Sharma

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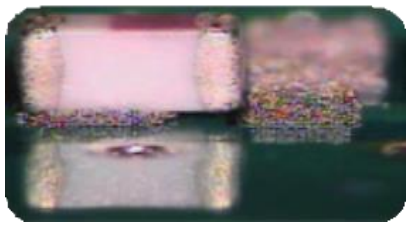
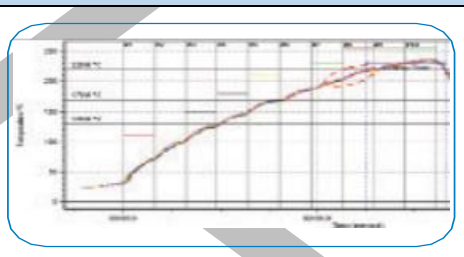
Skew placement will create imbalance wetting force on both pads.	Check other components placement accuracy. Re-teach fiducials if all component shifted, else edit that specific location manually.	

10.c) Possible Causes: Reflow

Description	Recommendations	Image
Extend soak zone can aid in balancing the wetting force on both pads before paste reaching to molten state	Focus at 30°C before alloy liquidus temperature, e.g., for SAC305, liquidus @ 220°C, ensure soak at 190~220°C for minimum of 30 seconds	

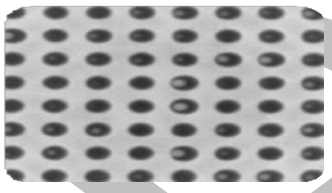
11. Voiding:

Definition: Voids in solder joints are empty spaces within the joint, increasing concern about voiding, especially on BGAs and large pads such as LGAs.

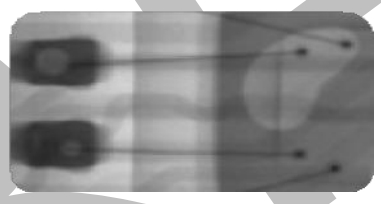
Two main contributors of voiding are

i)outgassing of flux entrapped

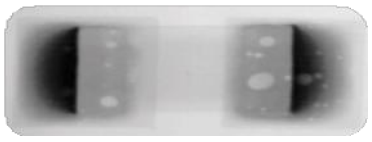
II)excessive oxidation.



BGA



LGA



Passive Component

11.a) Possible Causes: PCB

Description	Recommendations
Micro-via holes on pads trapped flux and air pockets	Typically, via holes <6mils will be more difficult to vaporize the flux or air trapped. Plug the blinded via holes before printing. Double print helps to pack more solder paste into via holes. Use finer powder size. Avoid printing paste on top of via holes, instead aperture opening designed around it.

11.b) Possible Causes: Stencil

Description	Recommendations

Review	Name
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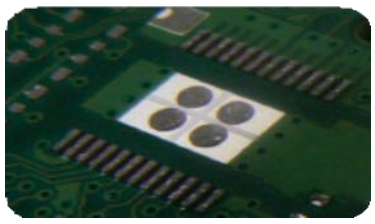
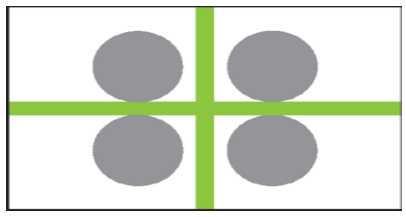
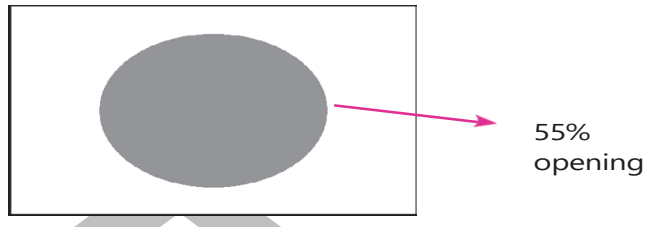
For large pads such as LGA, massive solder volume has a lot of flux to vaporize during reflow. Any trapped flux will result in voids.	Reduced amount of solder deposit Total solder volume reduction can be as high as 45%. With solder mask in between, break the large aperture into small openings. Without solder mask, cut a large round opening in the middle.

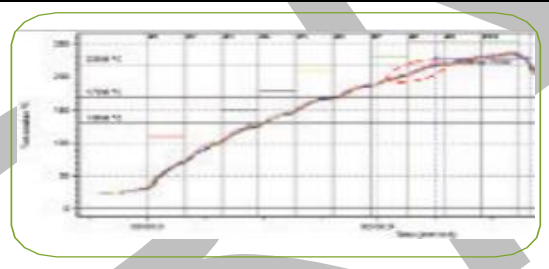






11.c) Possible Causes: Reflow Profile:

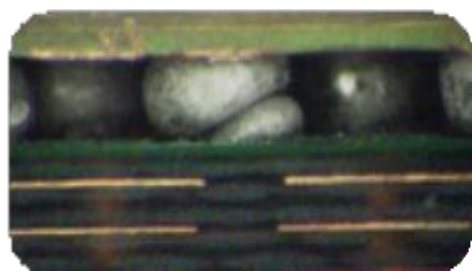
Description	Recommendations
Flux entrapped without sufficient time to outgas	Establish soak zone from 170~220°C for 60-90 sec. Also make sure profile set between 130~220°C for 180 sec.
Oxidation rate predominates	Adopt short profile concept to preserve flux activity, no soak zone. Use nitrogen if possible. Reduce the peak temperature to 241°C or below.



12. BGA (Ball Grid Array) Head-on-Pillow:

Definition: Head-on-pillow is an assembly defect in which the spheres from a BGA or CSP don't coalesce with the solder paste on the PCB pad. It is important to differentiate head-on-pillow from a defect caused simply by insufficient reflow temperature, which is characterized by distinct solder spheres from the paste that have not been properly melted on the pad and BGA solder sphere. With head-on-pillow the soldering

temperature is sufficient to fully melt the solder sphere and paste deposit, but an impediment to the formation of a proper solder joint exists.



12.a) Possible Causes: Screen Printer

Description	Recommendations	Image

Review	Name
Prepared By	Stanish Vijay
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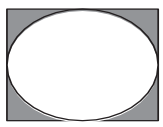
Irregular print definition across the pads may hinder some solder bump locations to be in contact with solder paste.	Verify print definition and measure print height consistency	

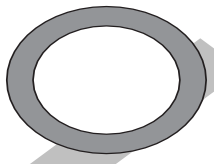
12.b) Possible Causes: PCB/Component

Description	Recommendations
Increase paste deposition volume to better compensate for substrate warpage.	Increase print volume by using square aperture vs. round opening, or enlarge overall deposition volume without jeopardizing bridging.



BGA coplanarity issue

- Oxidized BGA solder balls



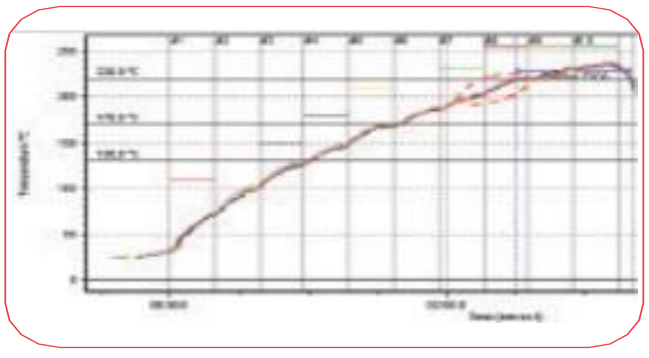
Increase solder volume.

- Use higher activity paste.
- Use nitrogen reflow.

12.c) BGA Head-on-Pillow:

Possible Causes: Reflow

Description	Recommendations
Board warpage especially for double reflow boards or thin PCBs (<1mm thick)	Recommendations Critical to minimize time above Tg, (typically 130°C for FR4 boards) with BGAs mounted. Target to maintain < 2 min if possible. For second reflow cycle, try to adopt lower preheat to reduce warpage occurrence.
Variance in CTE between PCB and BGA	Ensure minimum delta temperature difference between the BGA component and the rest of the components on the board. Apply short soak if necessary.
Paste hot-slump effect will aggravate BGA open joints if there are coplanarity issues.	Minimize time from 150°C to liquidus temperature.
Long soak profile may exhaust the flux capacity before reflow.	If a long soak is mandatory for complex board, use nitrogen cushion the flux capacity in overcoming oxidation rate.



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Approved By	Aditi Sharma

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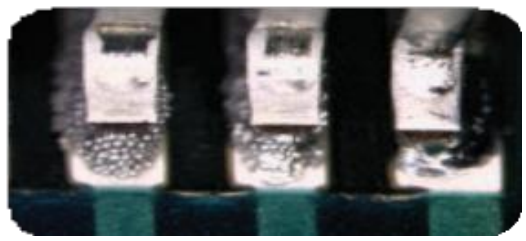
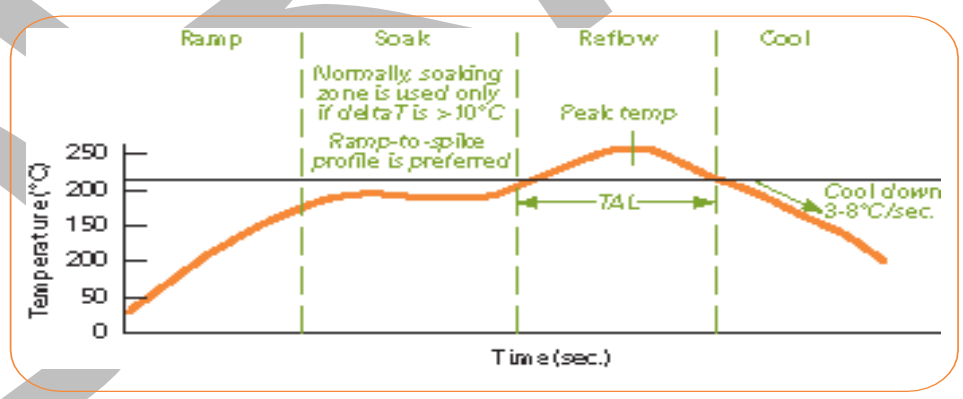
13. Grainy Joints

Definition: Sometimes called "Cold Solder," it is recognized by dark, non-reflective, rough surfaces from an alloy that is normally bright and shiny.



13.a) Possible Causes: Reflow

Description	Recommendations
Insufficient heat absorbed by the solder	Recommendations Ensure a TC is properly attached to this com- ponent. Verify peak temperature is at least 15°C above alloy liquidus and time above liquidus (TAL) > 45 sec.
Excessive heat imposed	Adopt a ramp-to-spike profile with soak zone to minimize oxidation and flux exhaustion. If soaking is mandatory, use nitrogen reflow whenever possible.
Cooling rate is too slow	Ensure alloy cooling rate from molten solder is 3-8°C/sec. Fast cooling rate will result in fine-grain structure appearance and looks shiny.



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